

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

Expected May 2028

College Park, MD

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Writing Java logic that sets the upper and lower bounds on EC2 capacity that Amazon Redshift reserves for operations like patching, balancing availability against cost, and backtesting it

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Rebuilt the legacy tool developers use to create test-data workflows as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation, and **90%** of 30+ surveyed developers preferred it
- Cut AI workflow generation time **65%** to under 15s by replacing a Claude skill's open-ended search with focused sub-skills that call Python tools for workflow search and validation, guided by CloudWatch and OpenSearch latency logs
- Because the 270K+ row PostgreSQL workflow table had strict access limits, precomputed popular workflows into a category tree that the Claude skill searches first, with Redis caching full workflows to cut database queries

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Deployed a space-topic research tool for Amazon Leo's public-policy team and set it up for handoff with AWS CDK infrastructure-as-code and GitHub Actions CI/CD, so Leo engineers new to the project now maintain it on AWS Fargate
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Automated retraining for a production Amazon Bedrock classifier that rates research documents by importance: on an EventBridge schedule a larger LLM grades its ratings, and fine-tuning runs only when the grades shift past a threshold

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- For a PhD-led robotics project aimed at ICLR, built a pipeline that overlays AnyGrasp's candidate grasps on camera images in Open3D for a vision-language model, then tuned prompts so the model picks the right grasp more often

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Co-founded a 110+ member student product studio that has shipped 5 live products
- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with 20,000+ visits in its first 30 days, to reduce database load and latency
- Building a RAG meal-suggestion feature for the official UMD app, in which GPT-4o mini writes suggestions from scraped recipes found by SentenceBERT vector search in ChromaDB, and internal testers rated **93%** of meals satisfactory

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine and replayed **50.9M** real IEX exchange messages through it without ever resyncing to the exchange's book, agreeing with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Implemented B+ tree and LSM-tree backends for the event log and benchmarked them on identical 2M-record workloads, where the B+ tree gave **12.8x** the read throughput and 22x faster range queries at the cost of 27% more disk space
- Sustained **1.26M events/sec** end-to-end over the full replay, keeping each price level's orders in a std::list FIFO inside a std::map and a hash map from order ID to list iterator for O(1) cancels

Technical Skills

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL, Bash

Cloud & DevOps: AWS (Lambda, Kinesis, Fargate, EC2, API Gateway, EventBridge, CloudWatch), AWS CDK (infrastructure as code), Docker, GitHub Actions (CI/CD)

Backend & Full-Stack: REST APIs, Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

AI/ML: LLMs (Claude, Bedrock, GPT-4o mini), retrieval-augmented generation (RAG), fine-tuning and evaluation, PyTorch

Practices & Tools: data structures and algorithms, unit testing (Pytest, Jest), profiling (perf, psutil), Linux, Git, Jira, Agile

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)